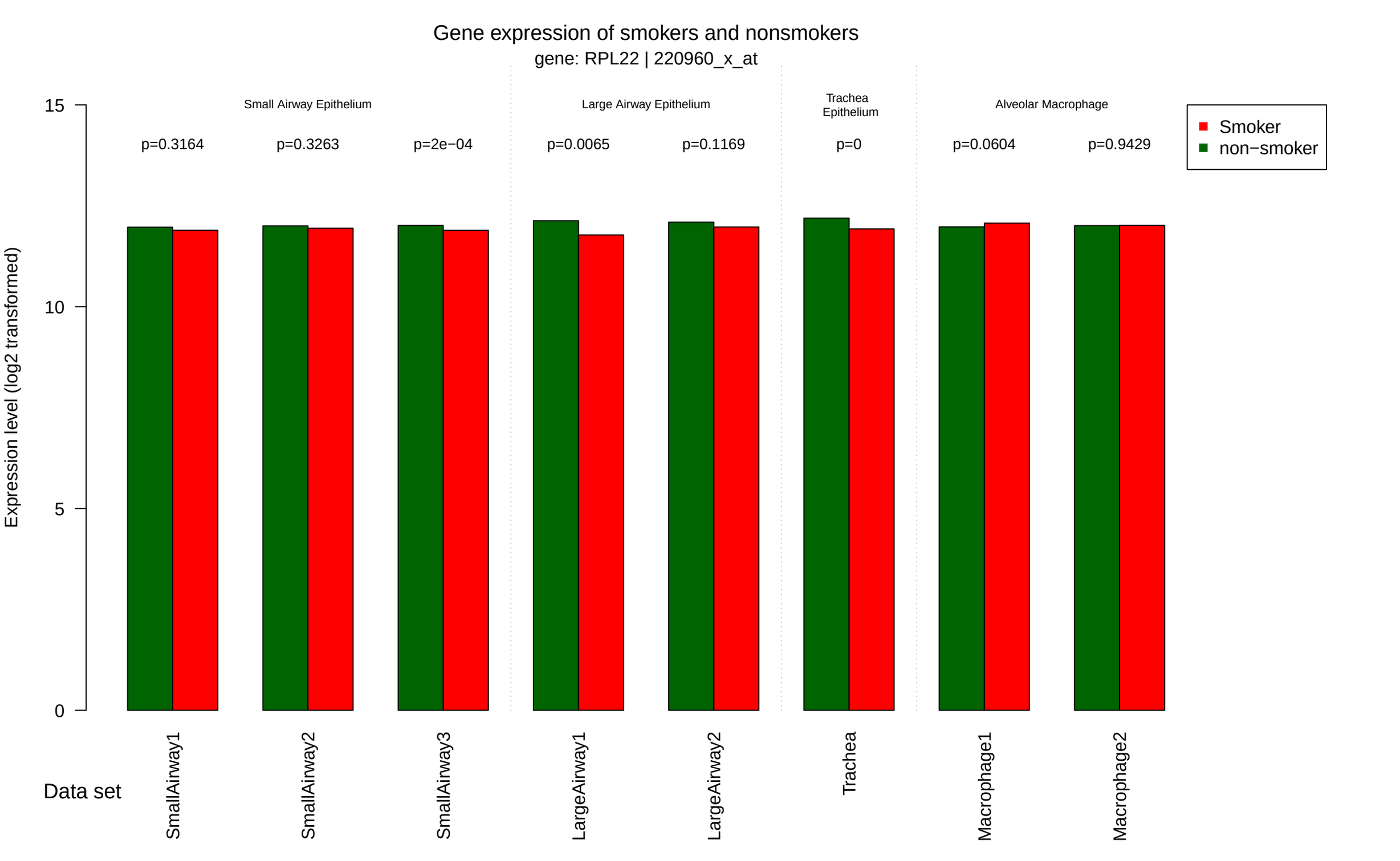
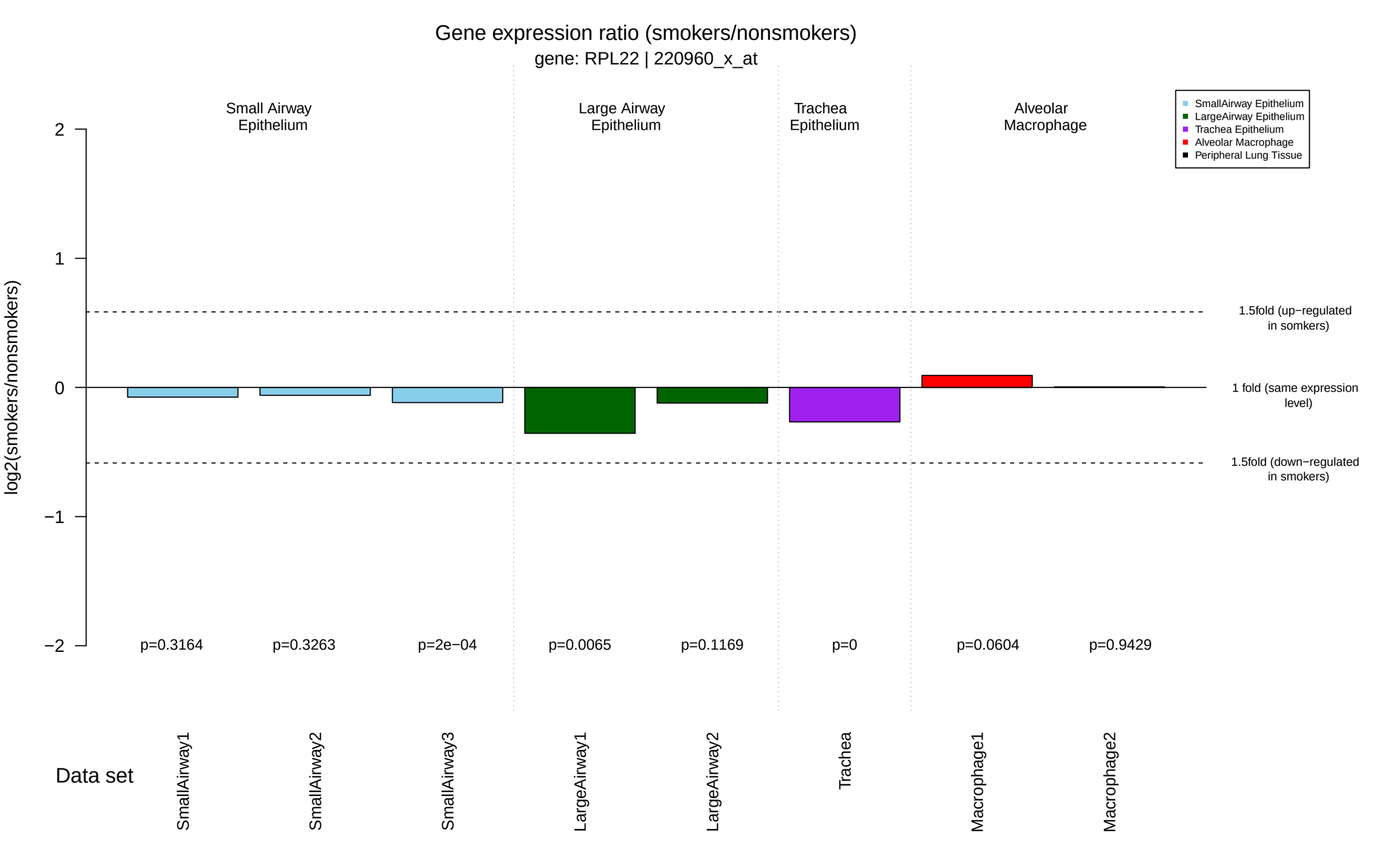
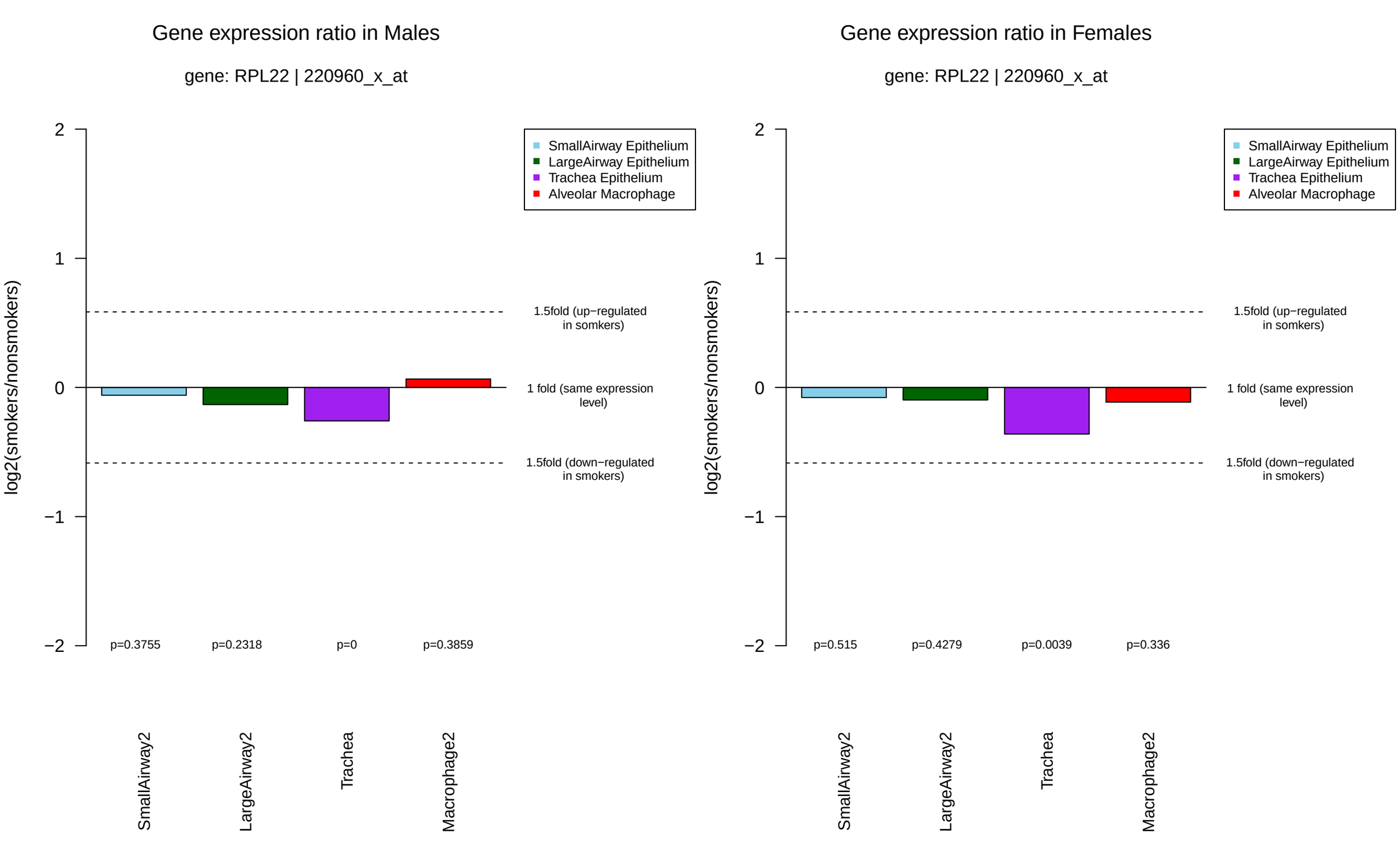
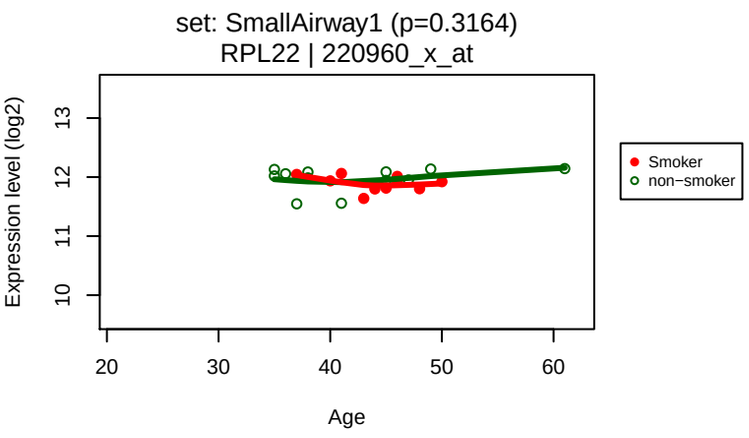
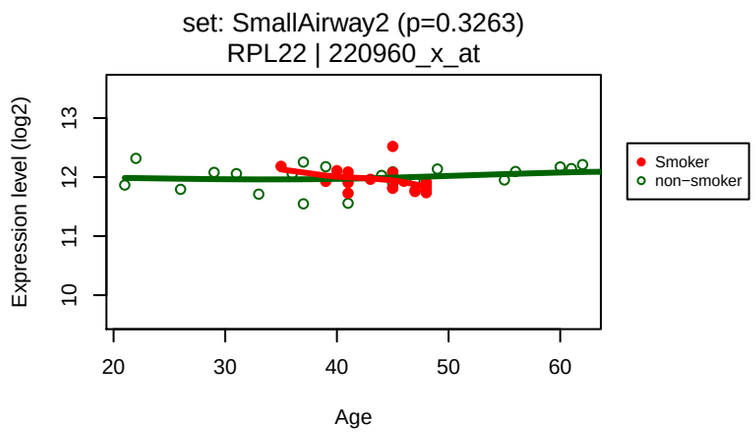
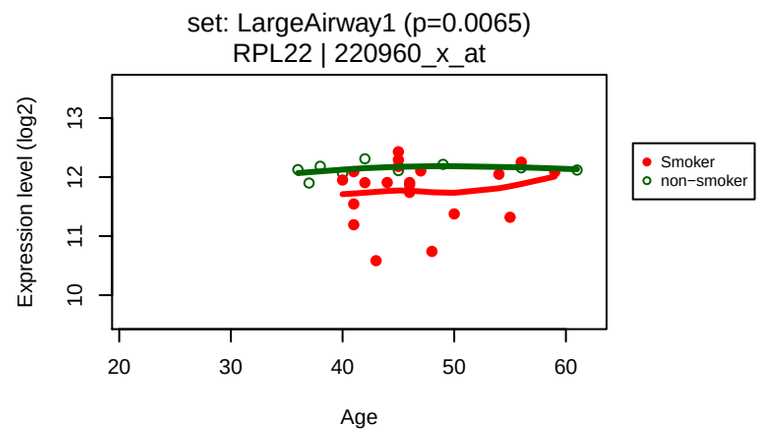

Gene expression of smokers and nonsmokers
gene: RPL22 | 220960_x_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.3164
p=0.3263
p=2e−04
p=0.0065
p=0.1169
p=0
p=0.0604
p=0.9429
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

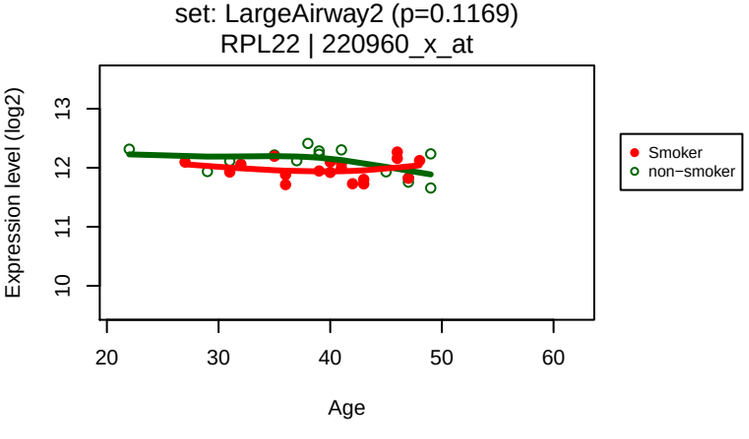

Gene expression ratio (smokers/nonsmokers)
gene: RPL22 | 220960_x_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.3164
p=0.3263
p=2e−04
p=0.0065
p=0.1169
p=0
p=0.0604
p=0.9429
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

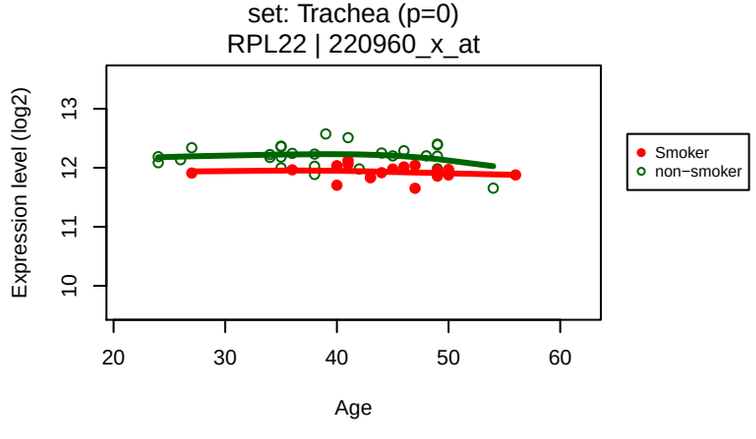

Gene expression ratio in Males
gene: RPL22 | 220960_x_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.3755
p=0.2318
p=0
p=0.3859
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: RPL22 | 220960_x_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.515
p=0.4279
p=0.0039
p=0.336
SmallAirway2
LargeAirway2
Trachea
Macrophage2

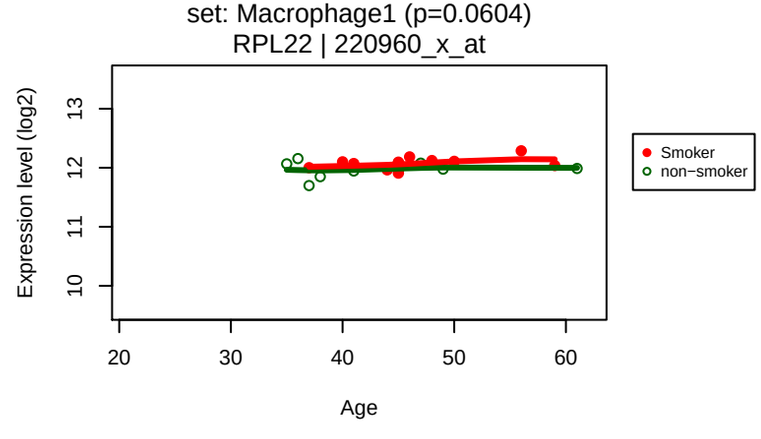

set: SmallAirway1 (p=0.3164)
RPL22 | 220960_x_at
Expression level (log2)
Age
Smoker
non−smoker

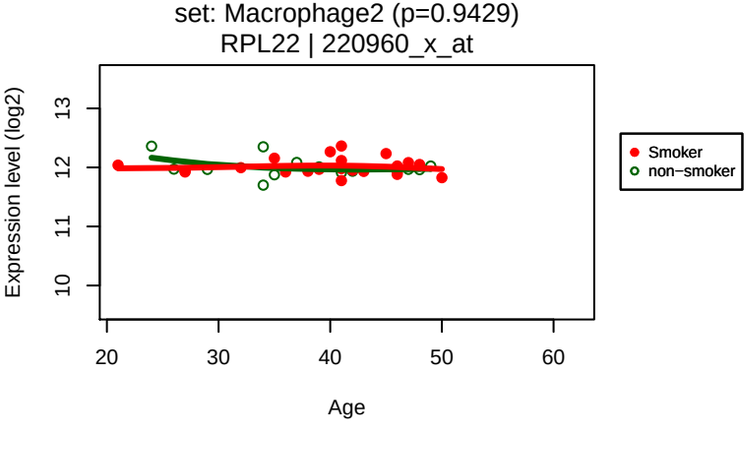

set: SmallAirway2 (p=0.3263)
RPL22 | 220960_x_at
Expression level (log2)
Age
Smoker
non−smoker

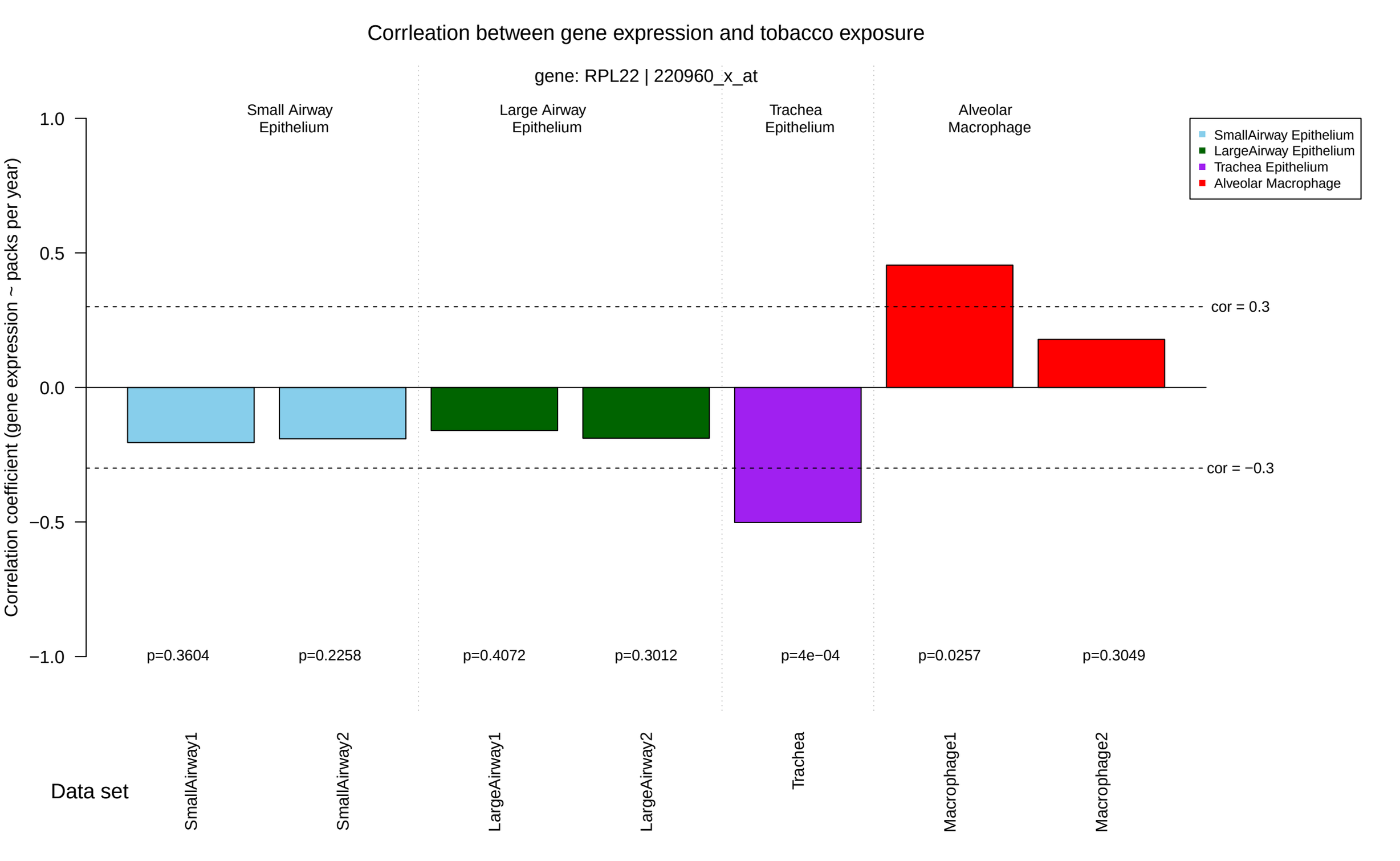

set: LargeAirway1 (p=0.0065)
RPL22 | 220960_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.1169)
RPL22 | 220960_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0)
RPL22 | 220960_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.0604)
RPL22 | 220960_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.9429)
RPL22 | 220960_x_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: RPL22 | 220960_x_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.3604
p=0.2258
p=0.4072
p=0.3012
p=4e−04
p=0.0257
p=0.3049
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2